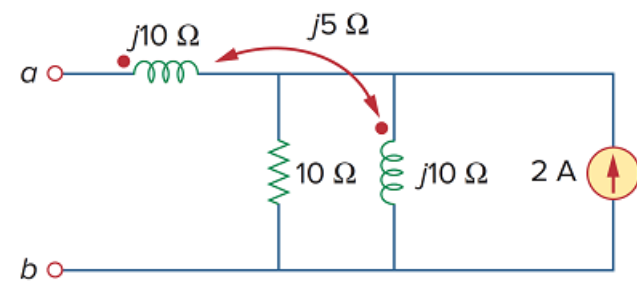


Problem

Find the Norton equivalent for the circuit in Fig. 13.84 at terminals a - b .

Figure 13.84



Step-by-step solution

Step 1 of 13

Refer to the Figure 13.84 from the textbook.

Calculate the Norton's impedance, Z_N .

Replace the current source by an open circuit and connect a 1 V source across terminals a and b . Assume the direction of loop currents I_1 and I_2 as shown in Figure 1.

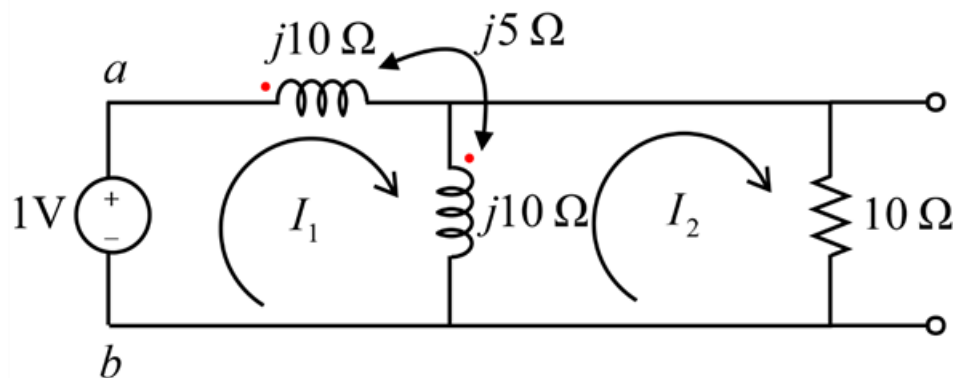


Figure 1

Step 2 of 13

Write the expression for the Norton's impedance, Z_N .

$$Z_N = \frac{1}{I_1}$$

Step 3 of 13

Write the expression for the mutual voltage, v_1 .

$$v_1 = j\omega M I_2$$

Since the current in the second coil is $(I_2 - I_1)$ as shown in Figure 1.

Thus, the mutual voltage v_1 due to current in second coil is,

$$v_1 = j5(I_2 - I_1)$$

Calculate the mutual voltage v_2 due to current in first coil.

$$v_2 = j5I_1$$

Step 4 of 13

Since the current I_2 is leaving the dot of the second coil, the polarity of the mutual voltage will be negative at the dot of the first coil.

Since the current I_1 is entering the dot of the second coil, the polarity of the mutual voltage will be positive at the dot of the first coil.

Replace the effect of mutual inductance by an equivalent voltage source of magnitude the same as the mutual voltage as shown in Figure 2.

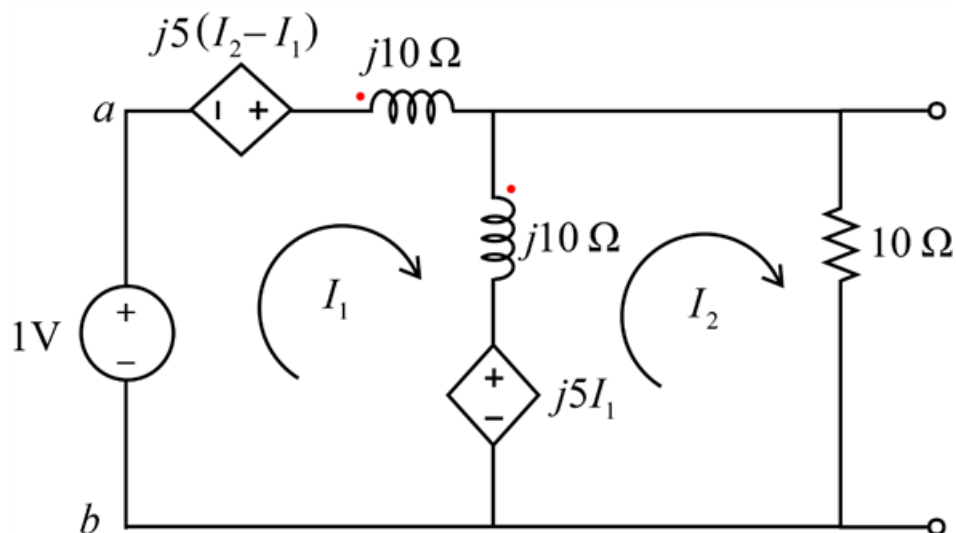


Figure 2

Step 5 of 13

Apply the Kirchhoff's Voltage Law in the loop with current I_1 .

$$1 = j10I_1 + j10(I_1 - I_2) - j5(I_2 - I_1) + j5I_1$$

$$1 = j10I_1 + j10I_1 - j10I_2 - j5I_2 + j5I_1 + j5I_1$$

$$1 = j30I_1 - j15I_2 \dots\dots (1)$$

Step 6 of 13

Apply the Kirchhoff's Voltage Law in the loop with current I_2 .

$$0 = j10(I_2 - I_1) + 10I_2 - j5I_1$$

$$0 = j10I_2 - j10I_1 + 10I_2 - j5I_1$$

$$I_2 = \frac{j15}{10 + j10} I_1$$

Step 7 of 13

Substitute $\frac{j15}{10 + j10} I_1$ for I_2 in equation (1).

$$1 = j30I_1 - j15\left(\frac{j15}{10 + j10} I_1\right)$$

$$1 = j30I_1 - \left(\frac{j^2 225}{10 + j10}\right) I_1$$

Substitute -1 for j^2 .

$$1 = j30I_1 - \left(\frac{(-1)225}{10 + j10}\right) I_1$$

$$1 = \left(\frac{j300 + j^2 300 + 225}{10 + j10}\right) I_1$$

$$1 = \left(\frac{-300 + 225 + j300}{10 + j10}\right) I_1$$

$$\frac{1}{I_1} = \frac{-25 + j300}{10 + j10}$$

[Comments \(1\)](#)

☐

Anonymous

-300+225= -75

Step 8 of 13

Substitute Z_N for $\frac{1}{I_1}$.

$$\begin{aligned} Z_N &= \frac{-75 + j300}{10 + j10} \\ &= 11.25 + j18.75 \, \Omega \end{aligned}$$

[Comments \(1\)](#)

☐

Anonymous

it should be $(-25+j300)/(10+j10)$ which comes out to $13.75+j16.25$

Step 9 of 13

Calculate the Norton's current, I_N .

Short-circuit terminals a and b . Since $20\ \Omega$ is in parallel with the current source, replace the current source by equivalent voltage source with $20\ \Omega$ in series with it as shown in Figure 3.

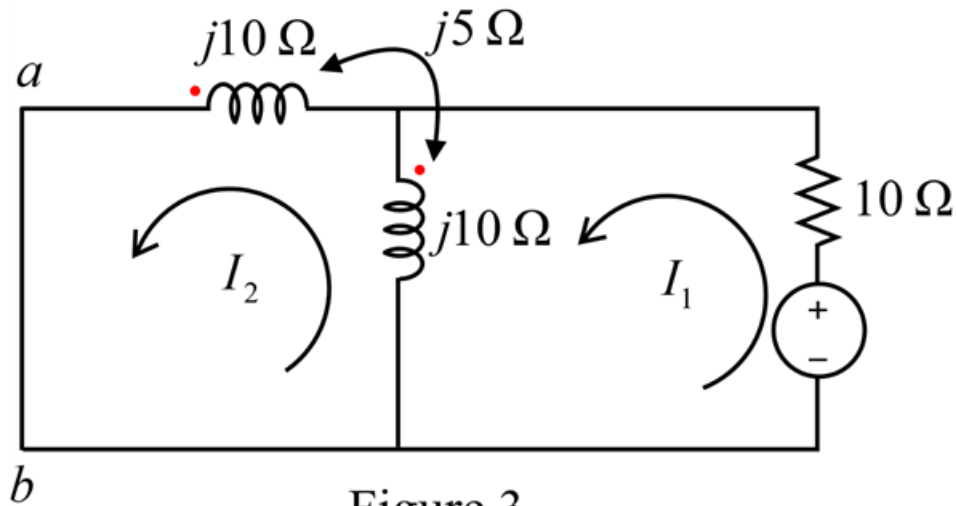


Figure 3

Step 10 of 13

Use dot convention and replace the effect of mutual inductance with corresponding mutual voltage.

Since the current I_2 is leaving the dot of the second coil, the polarity of the mutual voltage will be negative at the dot of the first coil.

Since the current $(I_1 - I_2)$ is entering the dot of the second coil, the polarity of the mutual voltage will be positive at the dot of the first coil.

Replace the effect of mutual inductance by an equivalent voltage source of magnitude the same as the mutual voltage as shown in Figure 4.

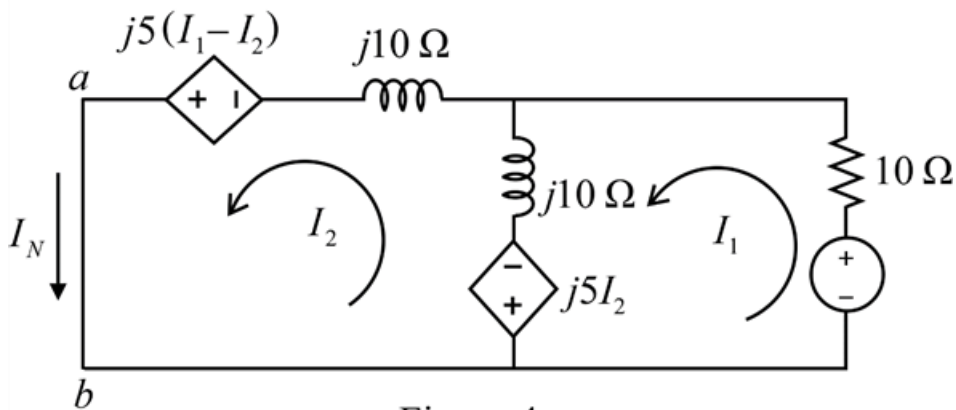


Figure 4

Step 11 of 13

Apply the Kirchhoff's Voltage Law in the loop with current I_1 .

$$20 = 10I_1 + j10(I_1 - I_2) - j5I_2$$

$$20 = (10 + j10)I_1 - j15I_2 \dots\dots (2)$$

Apply the Kirchhoff's Voltage Law in the loop with current I_2 .

$$0 = j10I_2 + j10(I_2 - I_1) - j5(I_1 - I_2) + j5I_2$$

$$0 = j30I_2 - j15I_1$$

$$I_1 = 2I_2$$

Step 12 of 13

Substitute $2I_2$ for I_1 in equation (2).

$$20 = (10 + j10)(2I_2) - j15I_2$$

$$20 = (20 + j5)I_2$$

$$I_2 = \frac{20}{20 + j5}$$

$$I_2 = 0.9701 \angle -14.036^\circ \text{ A}$$

Step 13 of 13

Since the short-circuit current is I_2 .

Thus,

$$I_N = I_2$$

$$= 0.9701 \angle -14.036^\circ \text{ A}$$

$$= 970.1 \angle -14.036^\circ \text{ mA}$$

Therefore, the norton's equivalent of the circuit will have a Norton's current source, I_N as

$970.1 \angle -14.036^\circ \text{ mA}$ and a Norton's impedance Z_N of $11.25 + j18.75 \Omega$.